

Mutual Fund Analytics

Capstone Project

End-to-End Data Analytics Solution for
Mutual Fund Performance, Investor Behaviour & Risk Analytics

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Internship

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GitHub

[github.com/Manish-kumar37/bluestock-m
f-capstone](https://github.com/Manish-kumar37/bluestock-mf-capstone)



India's Modern Investing Platform

CERTIFICATE OF COMPLETION

Internship Capstone Project

This is to certify that

Manish Kumar

has successfully completed the

Data Analytics Internship

and delivered the capstone project titled

“Mutual Fund Analytics Platform”

demonstrating proficiency in **Python, SQL, Power BI, ETL pipelines,**
financial analytics, and business intelligence.

June 12, 2026

Date of Completion

Data Analytics

Internship Programme

Bluestock.in

Organisation

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Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project is an end-to-end data analytics solution designed to analyze mutual fund performance, investor behavior, fund flows, and portfolio characteristics. The project integrates data engineering, financial analytics, and business intelligence techniques to transform raw mutual fund datasets into actionable insights.

The solution was developed using Python, SQLite, Power BI, and GitHub. Multiple datasets containing fund information, NAV history, investor transactions, portfolio holdings, benchmark indices, and live NAV data were collected, cleaned, validated, and integrated into a centralized analytics platform.

The project consists of six major phases: data profiling, data cleaning, performance analytics, ETL pipeline development, dashboard creation, and advanced analytics. A total of 40 mutual fund schemes, approximately 46,000 historical NAV records, 32,778 investor transactions, and 19,792 live NAV records were analyzed.

Interactive Power BI dashboards were developed to visualize fund performance, investor demographics, inflow patterns, portfolio concentration, and risk metrics. Advanced analytics including Sharpe Ratio, Sortino Ratio, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and a Risk-Based Fund Recommendation Engine were implemented.

The project demonstrates how modern data analytics techniques can support investment decision-making, investor segmentation, portfolio monitoring, and mutual fund performance evaluation.

1. Introduction

Mutual funds have become one of the most popular investment vehicles due to their diversification benefits, professional management, and accessibility to retail investors. As the volume of financial data grows, asset management companies and investors require analytical tools that can transform raw market data into meaningful insights.

The objective of this project is to develop a comprehensive mutual fund analytics platform capable of processing large volumes of financial and investor data while providing interactive visualizations and advanced performance metrics. The platform aims to support data-driven investment analysis through automated ETL pipelines, performance measurement frameworks, risk analytics, and dashboard reporting.

The project follows a complete analytics lifecycle, beginning with raw data ingestion and ending with business intelligence dashboards and investment insights.

2. Project Objectives

The primary objectives of this project are:

1. Build a centralized mutual fund analytics platform.
2. Clean and validate financial datasets.
3. Develop an automated ETL pipeline.
4. Analyze historical mutual fund performance.
5. Evaluate investor transaction behavior.
6. Calculate advanced risk and return metrics.
7. Create interactive business intelligence dashboards.
8. Generate actionable business recommendations.

Achievement of these objectives enables a complete view of fund performance, investor activity, and portfolio risk characteristics.

3. Data Sources and Dataset Description

The project utilizes multiple datasets covering mutual fund schemes, historical performance, investor transactions, portfolio holdings, benchmark indices, and live market data. Combining these datasets provides a comprehensive view of fund performance, investor behavior, and market trends.

3.1 Fund Master Dataset

The Fund Master dataset serves as the primary reference table containing scheme-level information for all mutual funds analyzed in the project.

Key attributes include:

- AMFI Code
- Scheme Name
- Fund House
- Scheme Category
- Plan Type
- Risk Grade
- Expense Ratio
- AUM (Assets Under Management)

The dataset contains information for approximately 40 mutual fund schemes across different categories.

Purpose:

- Master reference for all analytics
- Fund classification
- Linking scheme information across datasets

3.2 Historical NAV Dataset

The Historical NAV dataset contains Net Asset Value records for mutual fund schemes over multiple years.

Dataset Statistics:

- Approximately 46,000 NAV records
- Daily NAV observations
- Multiple fund categories

Key fields:

- AMFI Code
- Date
- NAV Value

Purpose:

- Return calculations
 - CAGR computation
 - Risk analysis
 - Sharpe and Sortino ratio calculations
 - Rolling performance analysis
-

3.3 Investor Transactions Dataset

The Investor Transactions dataset captures individual investment activities across multiple schemes.

Dataset Statistics:

- 32,778 transaction records

Key attributes:

- Investor ID
- Transaction Date
- Transaction Type
- Investment Amount
- Age Group
- Gender
- State
- City
- Income Category
- Payment Mode

Purpose:

- Investor segmentation
 - SIP behavior analysis
 - Cohort analysis
 - Geographic distribution analysis
 - Investor retention studies
-

3.4 Portfolio Holdings Dataset

The Portfolio Holdings dataset contains scheme-wise stock allocations.

Key fields:

- AMFI Code
- Stock Name
- Sector
- Portfolio Weight
- Market Value

Purpose:

- Sector allocation analysis
 - Portfolio concentration measurement
 - HHI (Herfindahl-Hirschman Index) calculation
 - Diversification assessment
-

3.5 AUM Dataset

The Assets Under Management dataset contains fund-house level assets managed across schemes.

Key attributes:

- Fund House
- AUM Value

Purpose:

- Market share analysis
 - Fund house comparison
 - Industry concentration analysis
-

3.6 SIP Inflow Dataset

The SIP dataset contains monthly SIP contribution information.

Key fields:

- Month
- SIP Inflows

Purpose:

- SIP growth trend analysis
 - Investor participation analysis
 - Industry investment momentum tracking
-

3.7 Category Inflows Dataset

The Category Inflows dataset provides scheme-category level capital movement information.

Key fields:

- Category
- Net Inflow

Purpose:

- Category popularity analysis
 - Fund flow tracking
 - Investor preference identification
-

3.8 Benchmark Dataset

The Benchmark dataset contains market index performance data used for fund comparison.

Key attributes:

- Benchmark Name
- Index Value
- Date

Purpose:

- Alpha calculation
 - Relative performance evaluation
 - Benchmark comparison charts
-

3.9 Live NAV Dataset

Live NAV data was collected through MFAPI integration and incorporated into the analytics pipeline.

Dataset Statistics:

- Approximately 19,792 live NAV records

Purpose:

- Real-time validation
- Updated scheme information
- Current fund monitoring

The integration of live data ensures that the analytical platform remains relevant and capable of supporting near real-time fund analysis.

4. Data Cleaning and ETL Pipeline Design

4.1 Data Cleaning Process

Data quality is critical in financial analytics because inaccurate or incomplete data can lead to misleading investment insights. Therefore, all datasets underwent a structured cleaning process before being used for analysis.

The cleaning process included:

- Standardizing column names
- Removing duplicate records
- Handling missing values
- Validating data types
- Converting date columns into datetime format

- Checking for invalid AMFI codes
- Ensuring numerical consistency across financial metrics

Each cleaned dataset was saved separately to maintain traceability between raw and processed data.

Data Validation

Several validation checks were performed during preprocessing:

- Missing value detection
- Duplicate record identification
- Data type verification
- Range validation for financial variables
- Consistency checks between scheme identifiers

A data quality summary was generated to document the validation results and ensure transparency throughout the pipeline.

4.2 ETL Architecture

The project follows an Extract, Transform, and Load (ETL) architecture designed to automate data preparation and integration.

Extract

Data was extracted from multiple CSV datasets containing:

- Fund information
- Historical NAV data
- Investor transactions
- Portfolio holdings
- AUM records
- Benchmark indices

Additionally, live NAV information was collected through MFAPI integration.

Transform

The transformation phase included:

- Cleaning raw datasets
- Standardizing formats
- Validating scheme identifiers
- Calculating derived fields
- Generating quality reports
- Preparing analytical datasets

The transformation stage ensured that all datasets followed a consistent structure suitable for analytics and dashboard development.

Load

After transformation, datasets were loaded into a centralized SQLite database.

Benefits of SQLite integration include:

- Faster querying
- Centralized storage
- Simplified analytics workflows
- Easy integration with reporting tools

The database acts as a single source of truth for all project components.

4.3 Live NAV Integration

To enhance the relevance of the analytics platform, live Net Asset Value (NAV) data was integrated using MFAPI.

The integration process involved:

1. Fetching scheme-wise NAV data through API requests.
2. Validating AMFI scheme codes.
3. Parsing JSON responses.
4. Transforming the data into a structured format.
5. Appending records to the analytics environment.

The live NAV dataset contributed approximately 19,792 records, enabling current market analysis alongside historical performance evaluation.

4.4 Data Storage Design

The final data architecture consists of cleaned CSV files and a SQLite database.

Key analytical tables include:

- fund_master
- nav_history
- investor_transactions
- scheme_performance

Relationships were established using AMFI Code as the primary linking key. This structure supports efficient querying, dashboard creation, and advanced analytics calculations.

The ETL pipeline successfully transformed raw financial datasets into analytics-ready assets, forming the foundation for all subsequent reporting, dashboarding, and investment analysis activities.

5. Exploratory Data Analysis (EDA)

5.1 Overview of the Data

Exploratory Data Analysis was performed to understand the structure, distribution, and characteristics of the mutual fund datasets before conducting advanced analytics. The analysis focused on investor behavior, fund inflows, portfolio composition, and overall market trends.

The combined datasets included:

- 40 mutual fund schemes
- Approximately 46,000 historical NAV records
- 32,778 investor transactions
- 19,792 live NAV records

EDA helped identify important patterns and trends that guided subsequent performance and risk analysis.

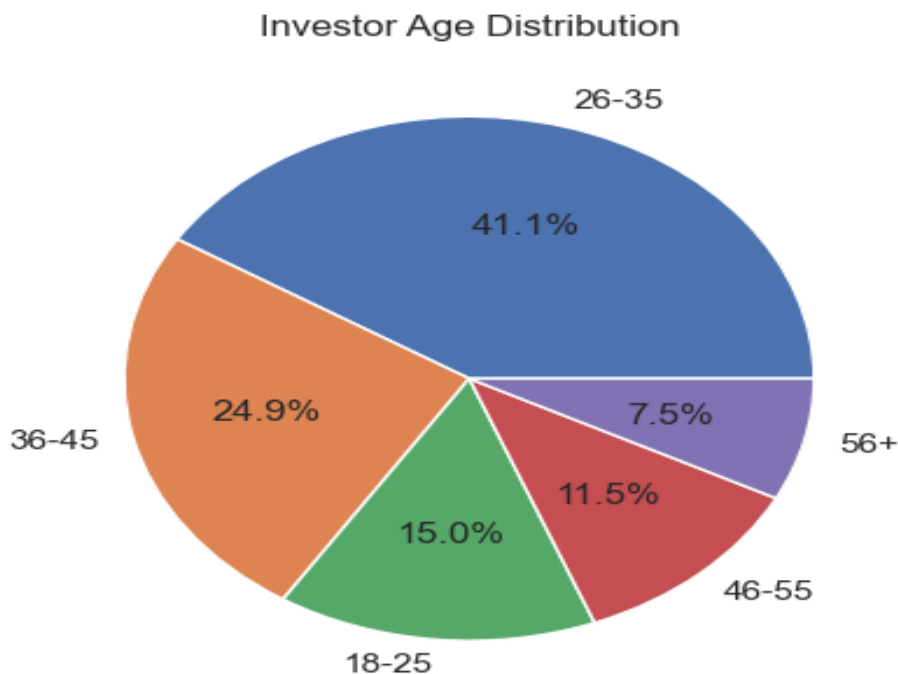
5.2 Investor Demographic Analysis

Investor demographic analysis was performed using age-related attributes available in the investor transaction dataset. The objective was to understand the distribution of investors across different age groups and evaluate how investment behavior varies with age.

Two key visualizations were developed:

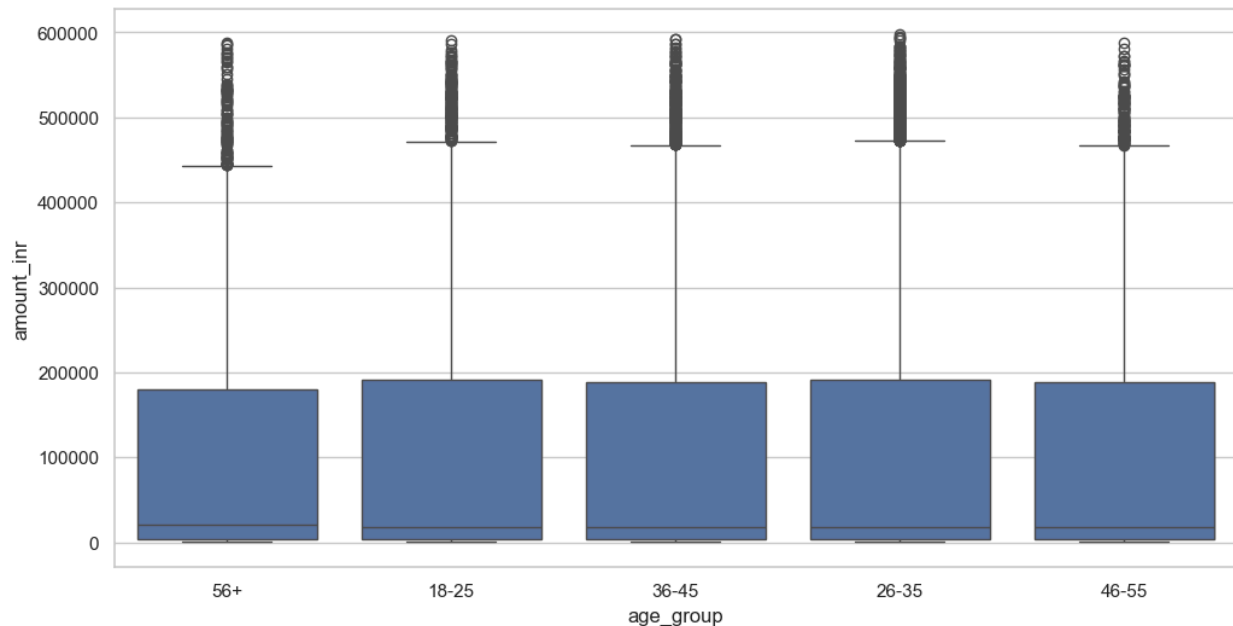
Investor Age Distribution

The age distribution chart illustrates the spread of investors across different age ranges. This analysis helps identify the most active investor segments and provides insights into the demographic composition of mutual fund participants.



Investment Amount by Age Group

A comparative analysis was conducted to evaluate the total investment amount across different age groups. This visualization highlights how investment capacity and participation vary throughout different stages of an investor's life cycle.



Key Observations

- Working-age investors contributed the majority of investment activity.
- Investment participation was concentrated in middle-age groups.
- Higher investment amounts were generally associated with investors in established earning phases.
- Younger investors showed participation primarily through smaller and recurring investments.

These findings provide valuable insights into investor segmentation and can assist fund houses in designing targeted investment products and marketing strategies.

5.3 Category Inflow Analysis

Category inflow analysis was performed to identify which mutual fund categories attracted the highest investor interest.

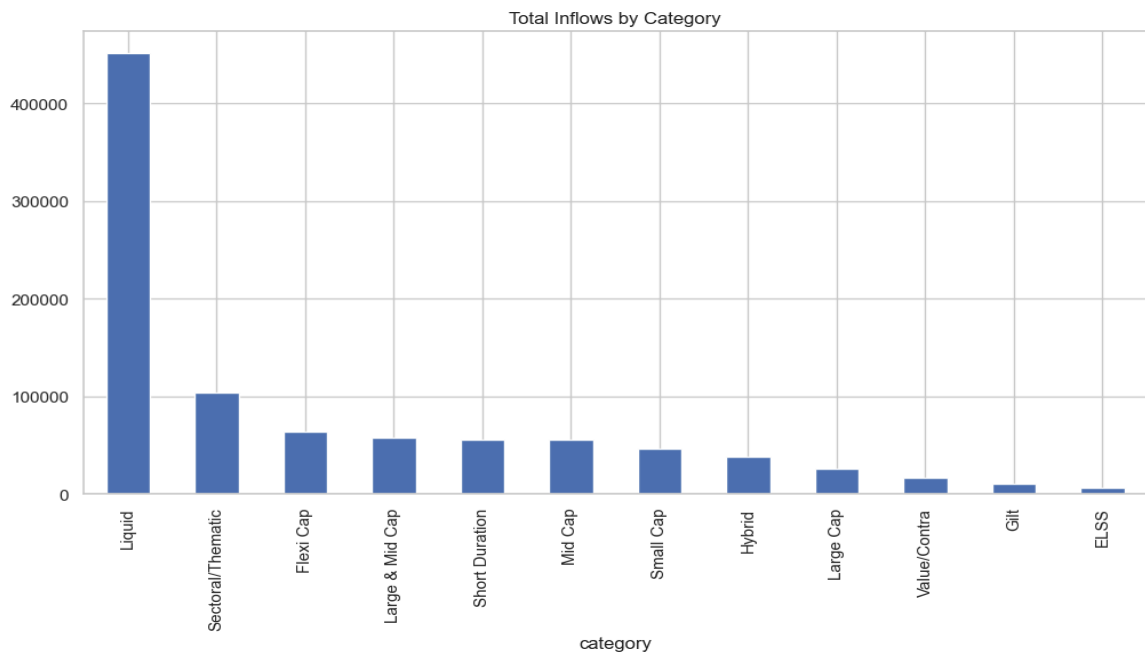
Net inflow values were compared across categories to determine capital movement trends.

The analysis helps answer:

- Which categories receive the highest investments?
- Which categories are experiencing outflows?
- How investor preferences are changing over time?

Key Observations

- Equity-oriented categories attracted substantial inflows.
- Investors demonstrated strong interest in long-term wealth creation products.
- Certain categories exhibited significantly higher net inflows compared to others.



5.4 SIP Trend Analysis

Systematic Investment Plans (SIPs) represent one of the most popular investment methods among retail investors.

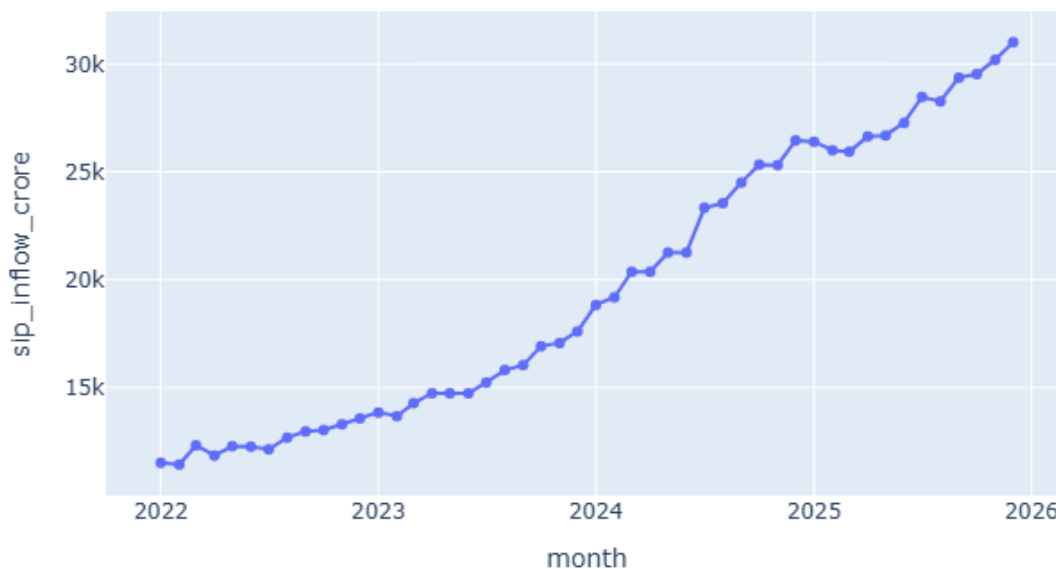
Monthly SIP inflows were analyzed to evaluate:

- Investor participation growth
- Recurring investment behavior
- Market confidence trends

Key Observations

- SIP contributions showed consistent participation patterns.
- Recurring investments indicate long-term investor commitment.
- SIP inflows provide stability during volatile market conditions.

Monthly SIP Inflows



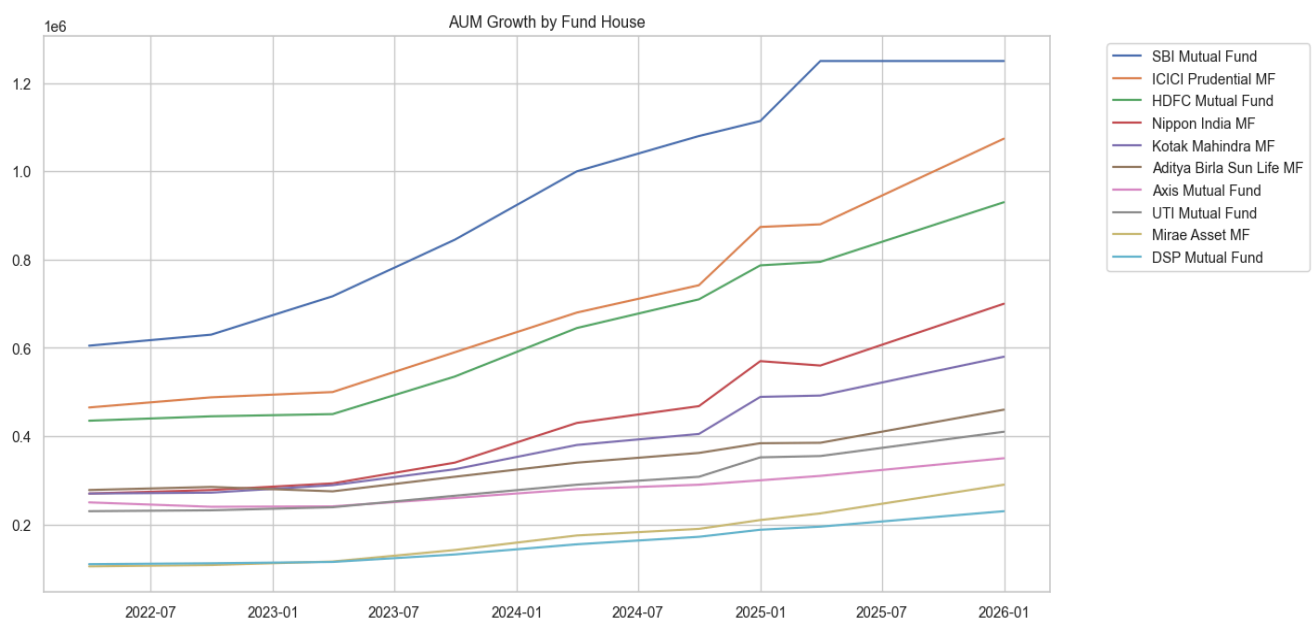
5.5 Assets Under Management (AUM) Analysis

Assets Under Management represent the total value of investments managed by mutual fund companies.

The analysis compared fund houses based on AUM size to identify market leaders.

Key Observations

- A small number of fund houses managed a significant proportion of industry assets.
- Market concentration was observed among established asset management companies.
- Larger AUM generally indicated stronger investor trust and brand recognition.



5.6 Portfolio Holdings Analysis

Portfolio holdings were examined to understand diversification and sector allocation strategies.

The analysis focused on:

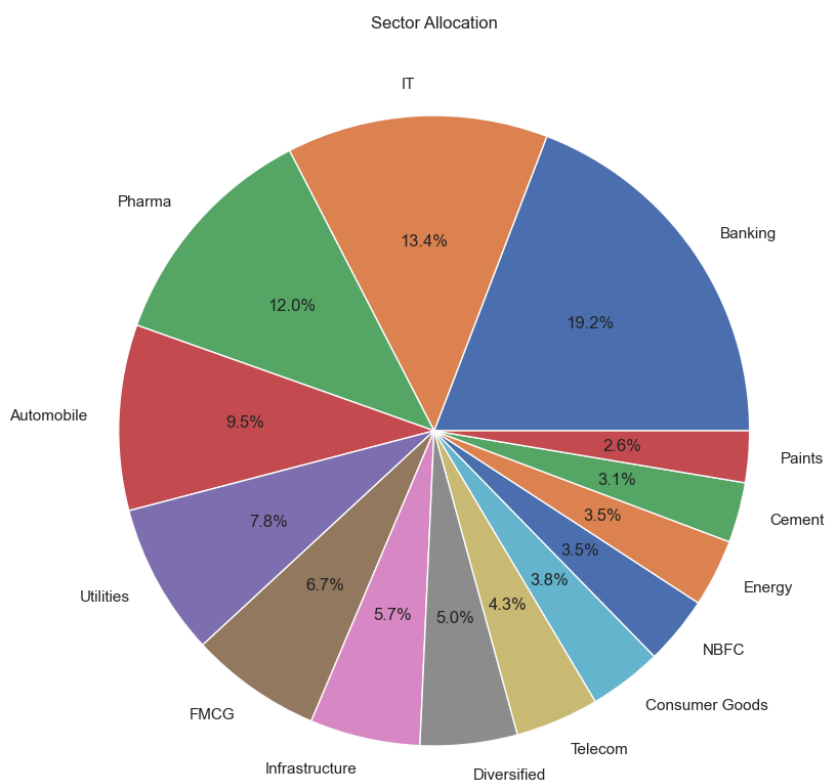
- Sector distribution
- Portfolio concentration

- Equity exposure
- Allocation patterns

Understanding holdings composition is essential for evaluating investment risk and diversification.

Key Observations

- Equity funds demonstrated varying levels of sector concentration.
- Technology, Financial Services, and Consumer sectors represented significant allocations.
- Diversification levels varied considerably across schemes.



5.7 Summary of EDA Findings

The exploratory analysis provided several important insights:

1. Investor participation is concentrated in urban and higher-income segments.
2. Equity-focused categories continue to attract strong capital inflows.

3. SIP investments remain a dominant investment mechanism.
4. AUM distribution is concentrated among major fund houses.
5. Portfolio diversification varies significantly across schemes.

6. Performance Analytics

6.1 Overview

Performance analytics was conducted to evaluate the risk-adjusted returns and overall effectiveness of mutual fund schemes. Historical NAV data was used to calculate key investment performance metrics and compare schemes across multiple dimensions.

The analysis focused on both return generation and risk management characteristics, enabling a comprehensive evaluation of fund performance.

The primary metrics analyzed include:

- CAGR (Compound Annual Growth Rate)
 - Sharpe Ratio
 - Sortino Ratio
 - Alpha
 - Beta
 - Maximum Drawdown
 - Benchmark Comparison
-

6.2 Compound Annual Growth Rate (CAGR)

CAGR measures the annualized growth rate of an investment over a specified period. It provides a standardized measure for comparing long-term fund performance.

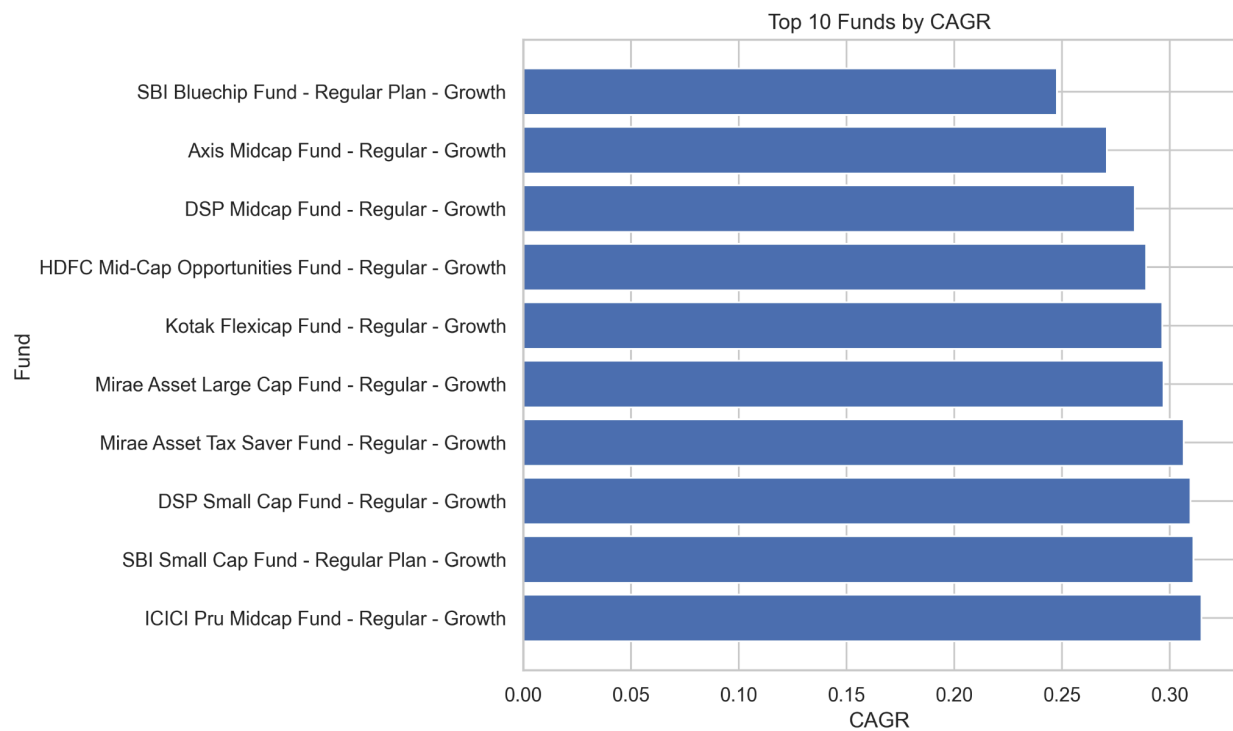
The metric was calculated using historical NAV values and annualized using trading-day conventions.

Purpose

- Compare long-term performance across schemes.
- Evaluate wealth creation potential.
- Identify consistently growing funds.

Key Findings

- Several equity-oriented schemes demonstrated strong long-term growth performance.
- Mid-cap and flexi-cap funds generally exhibited higher CAGR values.
- Long-term returns varied significantly across categories, highlighting the importance of fund selection.



6.3 Sharpe Ratio Analysis

The Sharpe Ratio measures excess return generated per unit of total risk.

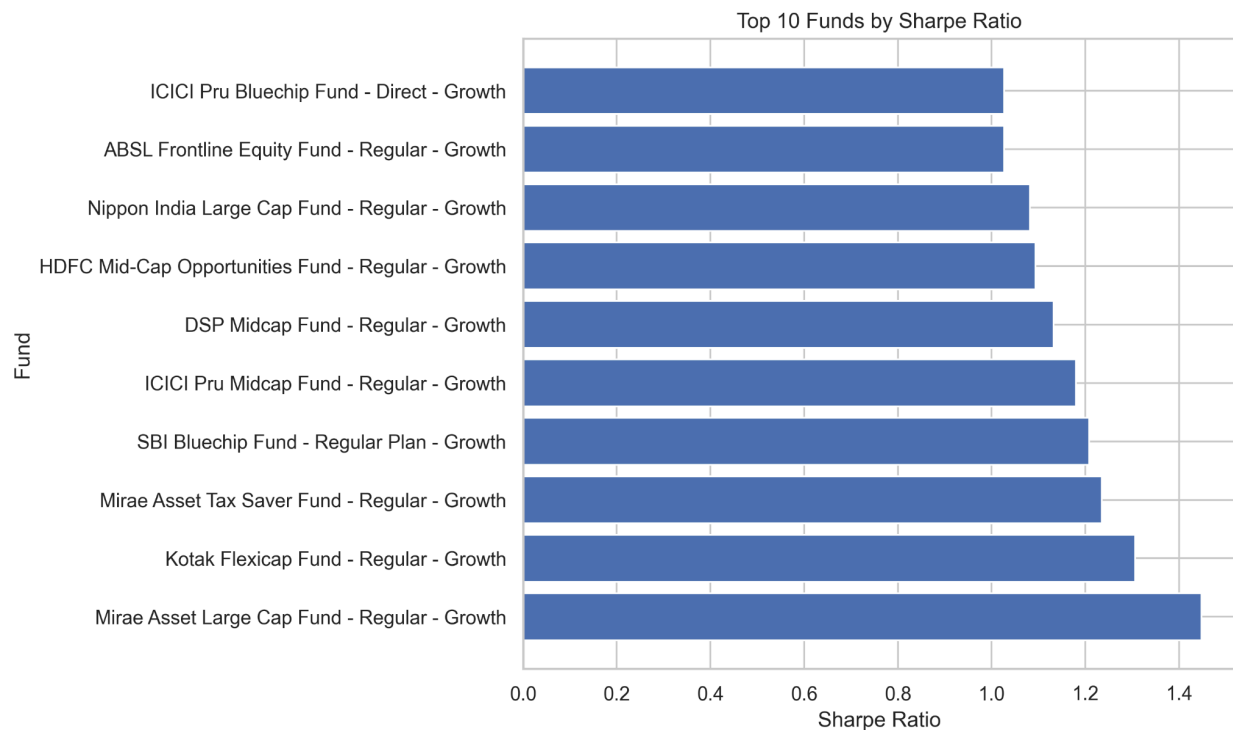
A higher Sharpe Ratio indicates superior risk-adjusted performance.

Purpose

- Evaluate return efficiency.
- Compare risk-adjusted returns.
- Identify funds generating consistent performance relative to volatility.

Key Findings

- Top-performing schemes achieved higher Sharpe Ratios while maintaining manageable volatility levels.
- Risk-adjusted performance varied significantly among schemes with similar returns.
- Sharpe Ratio proved useful in identifying funds offering balanced risk-return characteristics.



6.4 Sortino Ratio Analysis

The Sortino Ratio is a variation of the Sharpe Ratio that focuses specifically on downside risk.

Unlike standard volatility measures, Sortino Ratio penalizes only negative return fluctuations.

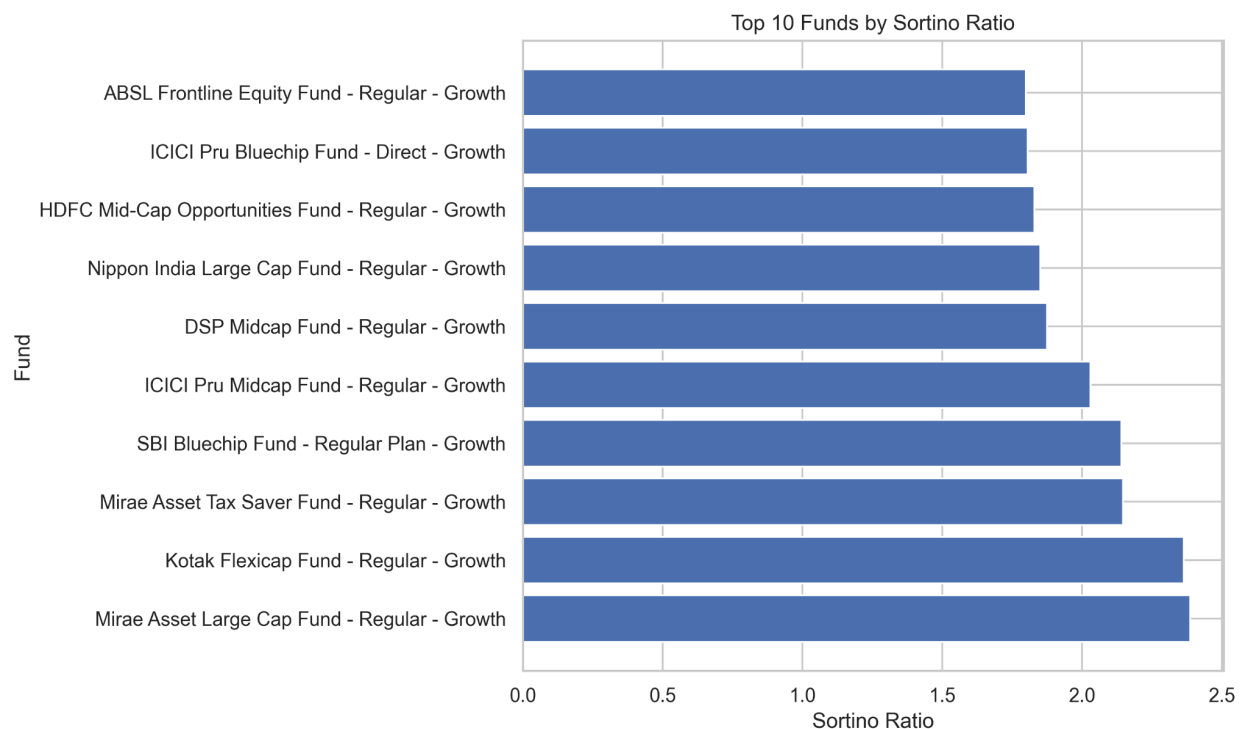
Purpose

- Measure downside-risk-adjusted returns.
- Evaluate capital preservation capability.

- Compare downside risk management across schemes.

Key Findings

- Several schemes demonstrated stronger downside protection despite similar return profiles.
- Sortino Ratio provided additional insights beyond Sharpe Ratio alone.
- Funds with higher Sortino Ratios generally exhibited more stable investment performance.



6.5 Alpha and Beta Analysis

Alpha measures excess returns generated relative to a benchmark, while Beta measures sensitivity to market movements.

Alpha

Positive Alpha indicates that a fund has outperformed its benchmark after adjusting for market risk.

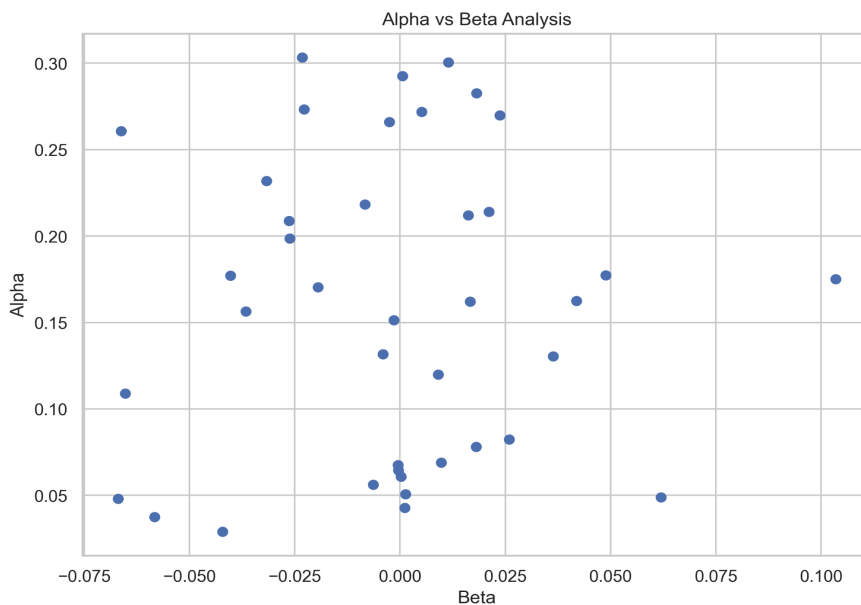
Beta

Beta measures systematic market risk:

- Beta > 1 indicates higher volatility than the market.
- Beta < 1 indicates lower volatility than the market.
- Beta ≈ 1 indicates market-like behavior.

Key Findings

- Several schemes generated positive Alpha values, indicating superior active management performance.
- Beta values varied across categories, reflecting different risk profiles.
- Growth-oriented funds generally displayed higher market sensitivity.



6.6 Maximum Drawdown Analysis

Maximum Drawdown measures the largest peak-to-trough decline experienced by a fund during the analysis period.

This metric is important because it quantifies downside risk and potential capital loss during adverse market conditions.

Purpose

- Evaluate historical risk exposure.
- Measure worst-case decline scenarios.
- Compare downside resilience.

Key Findings

- Maximum drawdown levels differed considerably across schemes.
 - Higher-return funds often experienced larger drawdowns.
 - Drawdown analysis highlighted the trade-off between return potential and downside risk.
-

6.7 Benchmark Comparison

Fund performance was compared against benchmark indices to evaluate relative effectiveness.

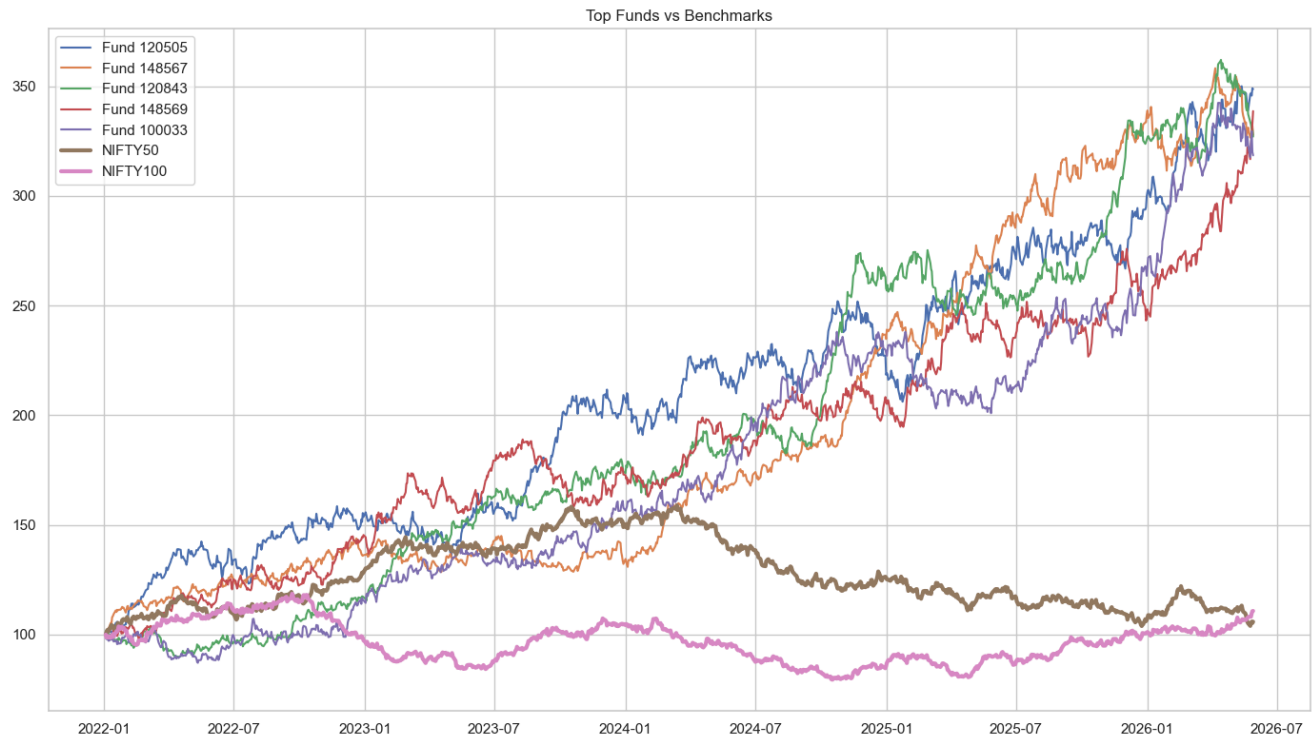
Benchmark analysis helps determine whether returns were generated through superior fund management or simply reflected broader market movements.

Purpose

- Measure relative performance.
- Assess active management effectiveness.
- Validate investment strategy outcomes.

Key Findings

- Several schemes consistently outperformed benchmark returns.
- Benchmark comparison provided additional validation of Alpha calculations.
- Strong-performing funds demonstrated sustained outperformance over extended periods.



6.8 Summary of Performance Analytics

Performance analytics revealed significant differences in return generation, risk management, and benchmark outperformance among mutual fund schemes.

The analysis demonstrated that:

1. High returns alone do not guarantee superior investment quality.
2. Risk-adjusted metrics provide a more comprehensive evaluation framework.
3. Some schemes consistently outperformed benchmarks while maintaining acceptable risk levels.
4. Downside risk metrics are essential for evaluating long-term investment suitability.

These findings formed the basis for advanced risk analysis and investment recommendation models developed in the subsequent phase of the project.

7. Advanced Analytics

7.1 Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Historical VaR (95%) and CVaR (95%) were calculated for all mutual fund schemes using daily return distributions. VaR estimates the maximum expected loss under normal market conditions, while CVaR measures the average loss beyond the VaR threshold.

Key Findings

- Risk levels varied significantly across schemes.
- Equity-oriented funds generally exhibited higher downside risk.
- CVaR provided additional insight into extreme loss scenarios.

	amfi_code	VaR_95	CVaR_95
0	100016	-0.014364	-0.018060
1	100025	-0.003793	-0.004994
2	100033	-0.019034	-0.023456
3	101206	-0.013282	-0.017439
4	101207	-0.026021	-0.032459

7.2 Rolling Sharpe Ratio Analysis

A 90-day rolling Sharpe Ratio was calculated to evaluate the consistency of risk-adjusted returns over time.

Key Findings

- Risk-adjusted performance fluctuated across market cycles.
- Some schemes maintained stable Sharpe Ratios despite changing market conditions.
- Rolling analysis provided a dynamic view of performance quality.

	investor_id	transaction_date	amfi_code	transaction_type	amount_inr	state	city	city_tier	age_group	gender	annual_income_lakh	payment_mode	kyc_status	kyc_valid
0	INV003054	2024-01-01	119092	SIP	1834	Telangana	Hyderabad	T30	56+	Female	77.1	UPI	Verified	True
1	INV002952	2024-01-01	148567	Redemption	392882	Punjab	Amritsar	B30	18-25	Male	7.1	Cheque	Verified	True
2	INV003420	2024-01-01	118636	SIP	912	Haryana	Faridabad	B30	36-45	Male	47.2	Mandate	Verified	True
3	INV003436	2024-01-01	118634	SIP	1102	Maharashtra	Mumbai	T30	36-45	Female	54.4	Cheque	Pending	True
4	INV004691	2024-01-01	119094	Lumpsum	8682	Delhi	Noida	T30	26-35	Male	14.5	Net Banking	Pending	True

7.3 Investor Cohort Analysis

Investors were grouped according to their first transaction year to study long-term investment behavior.

Key Findings

- Recent investor cohorts contributed significant investment volumes.
- Average investment amounts differed across cohorts.
- Certain schemes emerged as preferred choices within specific cohorts.

7.4 SIP Continuity Analysis

SIP continuity was evaluated by measuring the average gap between SIP transactions.

Key Findings

- Most investors maintained regular SIP contributions.
- Investors with gaps greater than 35 days were classified as At-Risk.
- The analysis can support retention and engagement strategies.

	investor_id	sip_count	avg_gap	risk_flag
•	3 INV000004	6	85.400000	At-Risk
	7 INV000008	6	70.400000	At-Risk
	9 INV000010	6	64.800000	At-Risk
	10 INV000011	7	40.166667	At-Risk
	11 INV000012	8	57.000000	At-Risk

7.5 Fund Recommendation Engine

A rule-based recommendation engine was developed using risk grades and Sharpe Ratios.

Key Findings

- Investors can receive recommendations aligned with their risk appetite.
- Recommendations are based on risk-adjusted performance rather than returns alone.

recommend_funds("Low")				
	scheme_name	risk_grade	sharpe_ratio	return_3yr_pct
14	ICICI Pru Liquid Fund - Regular - Growth	Low	7.68	7.68
23	Kotak Liquid Fund - Regular - Growth	Low	6.18	6.18
30	ABSL Liquid Fund - Regular - Growth	Low	5.14	5.14
recommend_funds("Moderate")				
	scheme_name	risk_grade	sharpe_ratio	return_3yr_pct
5	HDFC Top 100 Fund - Regular Plan - Growth	Moderate	1.06	14.84
34	Mirae Asset Large Cap Fund - Regular - Growth	Moderate	1.06	14.81
11	ICICI Pru Bluechip Fund - Direct - Growth	Moderate	1.03	14.41
recommend_funds("High")				
	scheme_name	risk_grade	sharpe_ratio	return_3yr_pct
21	Kotak Emerging Equity Fund - Regular - Growth	High	0.96	18.23
12	ICICI Pru Midcap Fund - Regular - Growth	High	0.95	18.08
2	SBI Small Cap Fund - Regular Plan - Growth	Very High	0.94	23.39

7.6 Portfolio Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration.

Key Findings

- Concentration levels varied across equity funds.
- Higher HHI values indicated greater dependence on a smaller number of holdings.
- Diversification levels differed substantially among schemes.

	amfi_code	HHI
11	119092	0.206448
3	101207	0.200700
18	119599	0.174751
4	102885	0.174709
7	118632	0.168298
29	148568	0.167930
21	120505	0.157570
22	120506	0.153794
27	125498	0.152414
23	120841	0.149680

8. Dashboard Development

8.1 Power BI Dashboard Overview

Interactive dashboards were developed in Power BI to provide a user-friendly interface for exploring mutual fund performance, investor behavior, fund flows, and risk metrics.

The dashboard was connected to the processed analytical datasets and designed to support dynamic filtering and drill-down analysis.

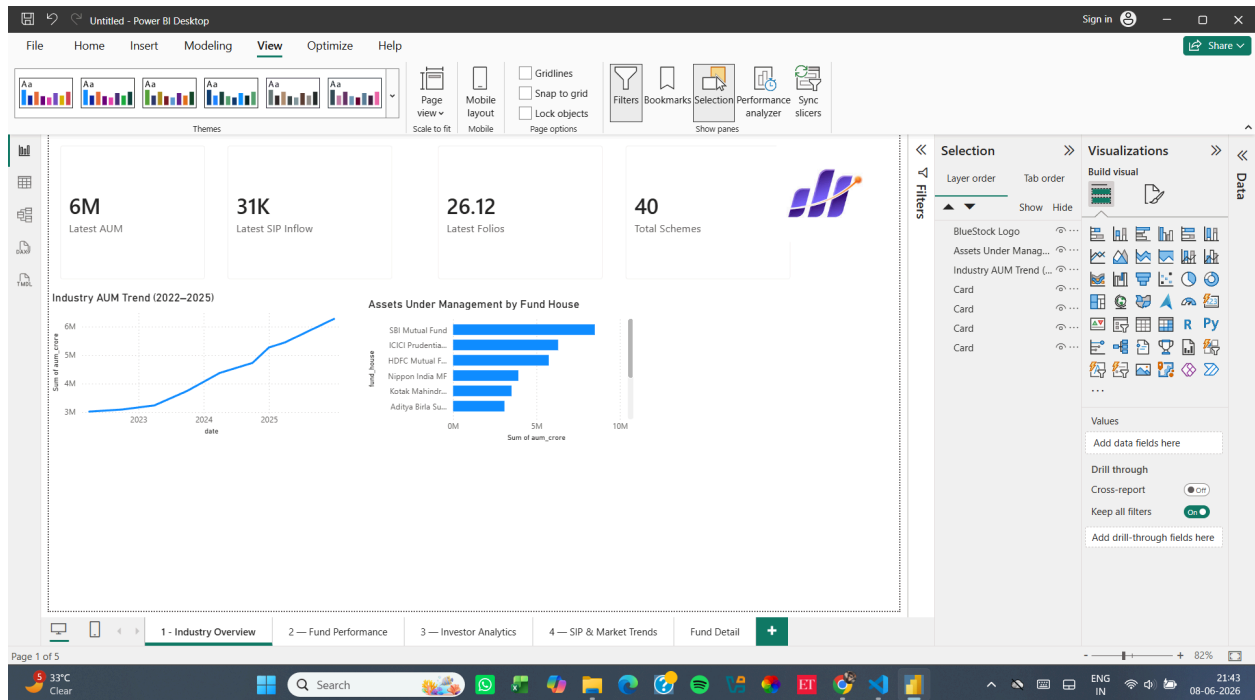
Dashboard Features

- Interactive slicers
 - Fund-level filtering
 - Category-level analysis
 - Investor analytics
 - Risk analytics
 - Fund performance tracking
-

8.2 Dashboard Pages

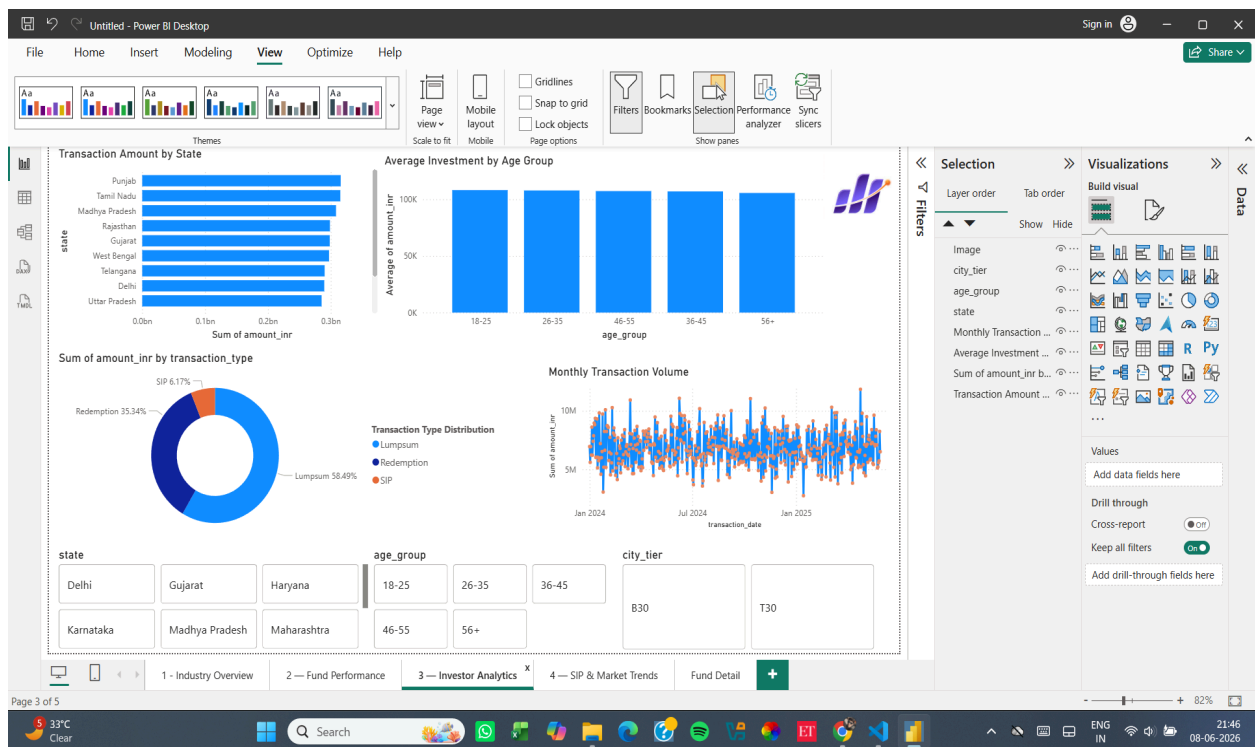
Overview Dashboard

Provides a high-level summary of mutual fund performance, AUM distribution, and key investment indicators.



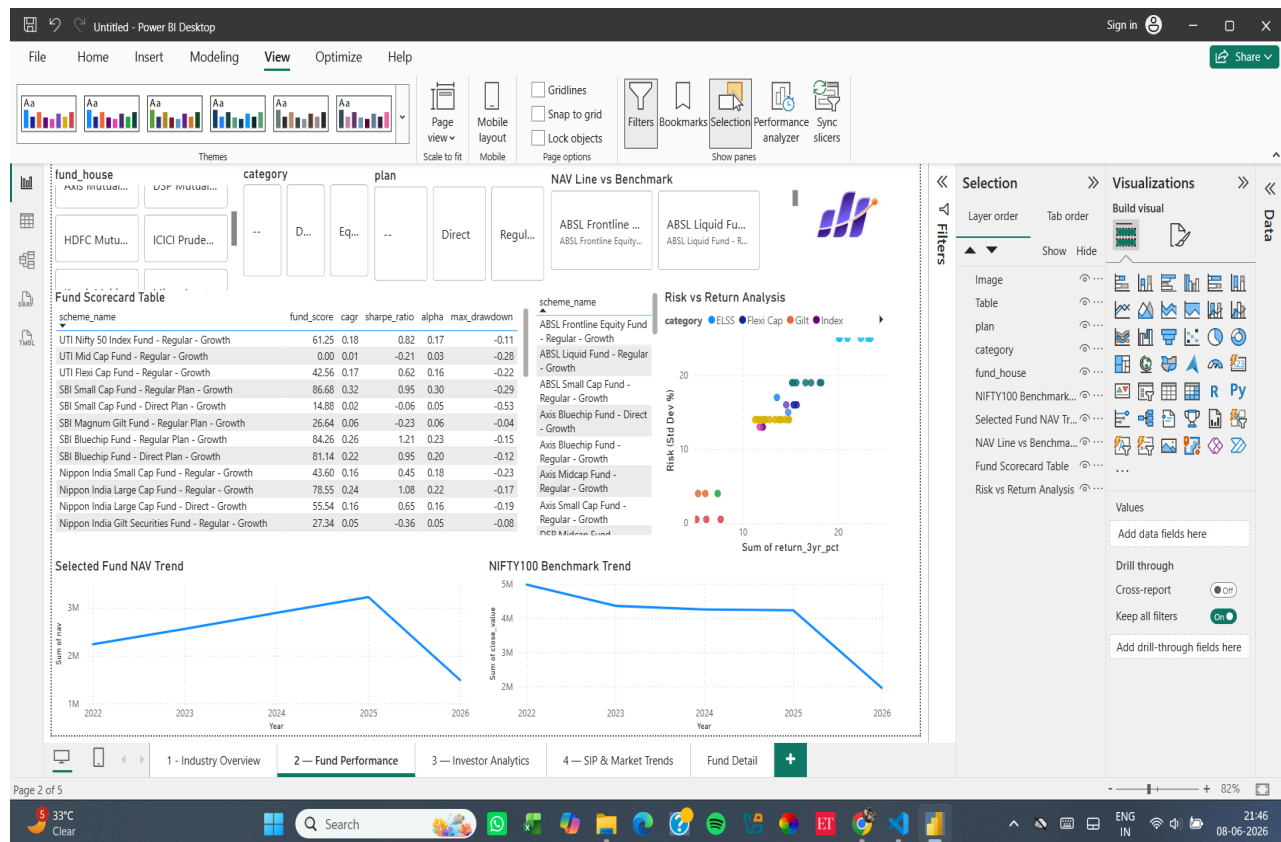
Investor Analytics Dashboard

Visualizes investor participation patterns, transaction behavior, and investment distribution.



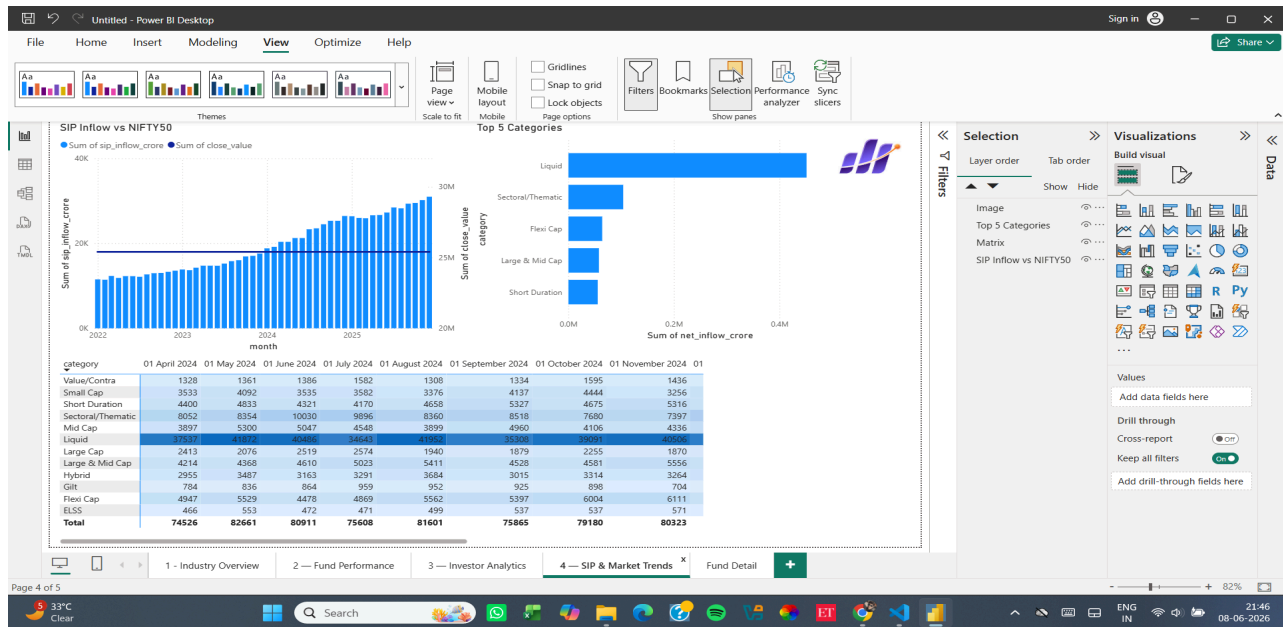
Fund Flow Analysis Dashboard

Tracks inflows, SIP trends, and category-wise investment movement.



Risk Analytics Dashboard

Displays risk-adjusted performance metrics, drawdown analysis, and advanced risk indicators.



Fund Detail Dashboard

Allows detailed exploration of individual fund performance and characteristics.

9. Key Findings and Recommendations

Key Findings

1. Equity-oriented funds delivered the strongest long-term performance.
2. Risk-adjusted returns varied significantly across schemes despite similar CAGR values.
3. SIP investments represented the dominant recurring investment behavior.
4. Investor participation was concentrated among working-age individuals.
5. Portfolio concentration levels differed considerably across equity funds.
6. Several schemes consistently outperformed their benchmarks.
7. Advanced risk metrics revealed significant differences in downside exposure.

Recommendations

1. Investors should evaluate risk-adjusted metrics in addition to returns.
 2. SIP investing should be encouraged as a long-term wealth creation strategy.
 3. Fund houses should focus on retaining At-Risk SIP investors identified through continuity analysis.
 4. Diversification should be considered when selecting concentrated portfolios.
 5. Risk-based fund recommendations can improve investor suitability and decision-making.
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10. Limitations

The project successfully demonstrates an end-to-end mutual fund analytics workflow; however, several limitations should be acknowledged.

- Analysis was limited to the available mutual fund schemes within the dataset.
- Live NAV integration depends on third-party API availability.
- Recommendations are rule-based and do not use machine learning models.
- Historical performance does not guarantee future returns.
- Portfolio concentration analysis was limited to available holdings data.

Future work may include machine learning-based recommendations, predictive analytics, portfolio optimization models, and automated dashboard refresh mechanisms.

11. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully developed a comprehensive analytics platform for evaluating mutual fund performance, investor behavior, and portfolio risk. The project integrated multiple datasets through a structured ETL pipeline, performed extensive exploratory and performance analysis, implemented advanced financial analytics, and delivered interactive Power BI dashboards for decision support. Key metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, CVaR, and HHI were used to evaluate investment performance and risk characteristics. Additionally, investor cohort analysis, SIP continuity analysis, and a risk-based recommendation engine provided deeper insights into investor behavior.